

Homework 10

STAT 300 (Fall 2024)

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Clearly label your answers to each question and each sub-question. Your answers MUST be uploaded to Canvas as a <HWx_Yourfirstname.nb.html> file by the deadline.

Automobile manufacturer is exploring designs for water pumps for their off-road vehicles. They have been looking into four prototypes designed by two companies. **Twelve** pumps of each design has been randomly installed and tested in four-wheel-drive vehicles and the number of miles driven before the failure of water pumps were recorded.

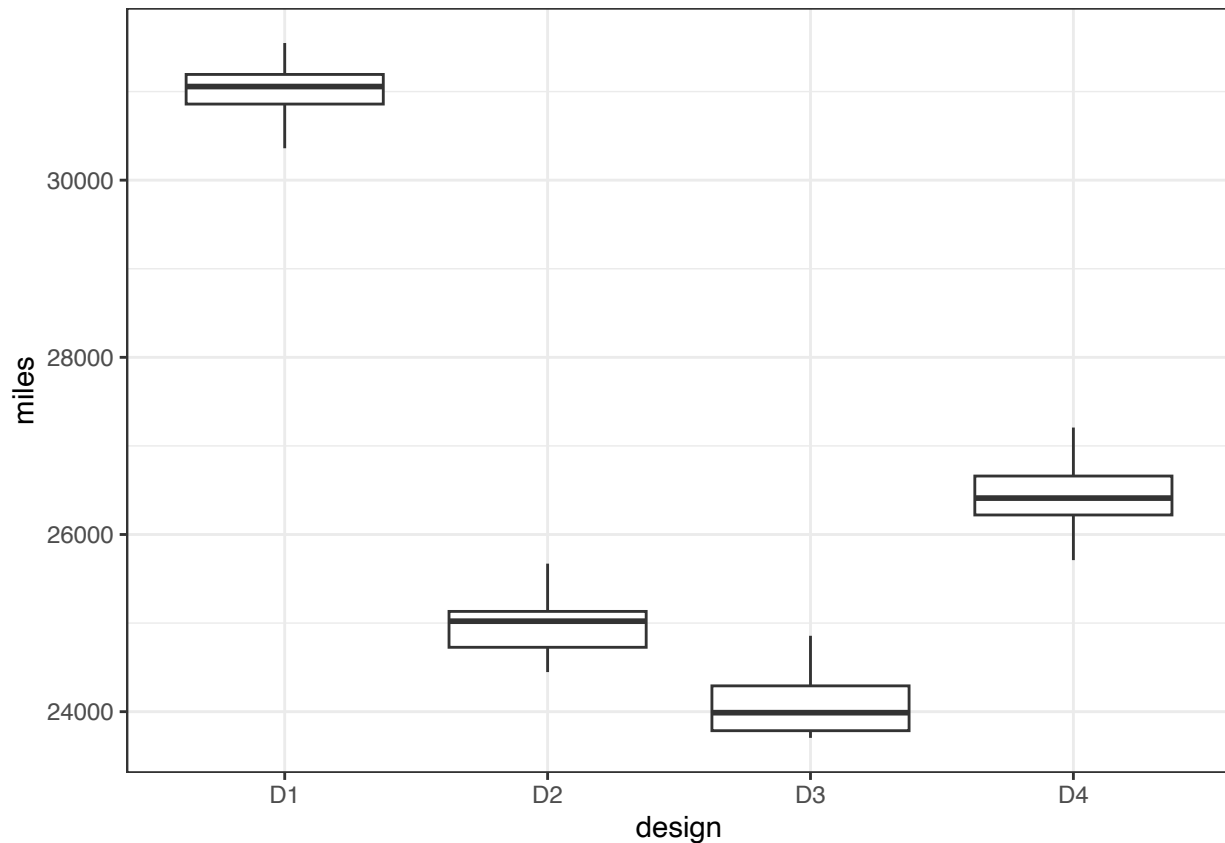
The company is specifically interested in getting answers to below research questions:

- I. Whether or not the average mileage until failure is different in four brands.
- II. Designs 1 and 2 were presented by company A whereas Designs 3 and 4 by company B. Is the average mileage at failure is differ by the company?
- III. Is the average mileage at failure is different for two designs, within each company?

From the methods we discussed in the class, think about how you would approach each of the above research questions.

Below is a set of questions that will help you to answer above research questions.

- (a). What is the response variable and what is the explanatory factor?
- (b). What are the treatments in this study?
- (c). Below provided is a graph related to the study. Comment on your observations.



(d). Whatever the differences you are observing above, your client wants to know if this simply could have happened by chance. To answer this, what is the model you think would explain the relationship between the two variables in the target population. Clearly define any terms you may have used.

(e). Carry out an appropriate test to see whether or not the average mileage until failure is different in four brands.

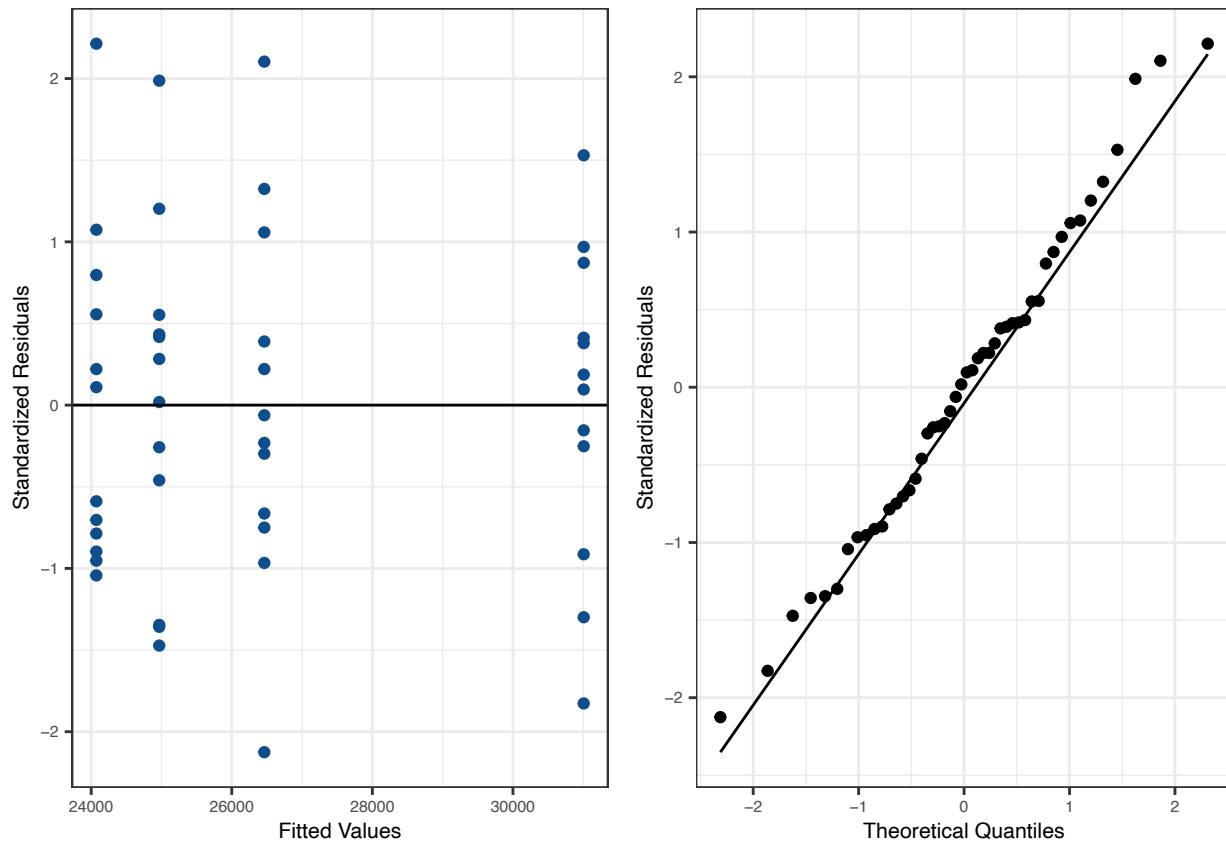
(i). Set up appropriate null and alternative hypothesis

(ii). Below provided is a partial ANOVA table. Use the information provided to you to complete the ANOVA table. Show your work to get the values in (A)-(H)

```
##           Df      Sum Sq   Mean Sq F value Pr(>F)
## design    (A) 341757430 113919143      (F)      (H)
## Residuals (B)      (D)      (E)
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Source	df	Sum of Squares	Mean Squares	F Test	P value
Group	(A)	341757430	(F)	(G)	(H)
Residual	(B)	(D)	136626		
Total	(C)	(E)			

(iii) Below are some graphs relevant as you check the assumptions. Use the provided graphs to comment on the assumptions.



- (iv) Regardless your opinion on the validity of the assumptions, write the conclusion in the context of the research question.
- (v) Which research question: I, II or III is answered by the test you conducted in questions (i) through (iv) in part (e)?

(f). Here are some outputs that will be helpful for answering research questions II and III above.

```
aggregate(miles~design,data=wpump,FUN=mean)
```

```
##  design  miles
## 1    D1 31006.51
## 2    D2 24968.01
## 3    D3 24073.46
## 4    D4 26462.88
```

```
qt(0.025,df=44)
```

```
## [1] -2.015368
```

```
qt(0.05/6,df=44)
```

```
## [1] -2.488968
```

```
qt(0.05/3,df=44)
```

```
## [1] -2.196979
```

- (i) Set up the correct contrasts relevant to answering research questions.
- (ii) Write down the sample contrasts/point estimators relevant for part (i)

- (ii) Calculate the *adjusted* CIs relevant for contrasts you set up above. Do the calculations by hand by typing up the work.
- (iii) Use the CIs you calculated in part (ii) to answer the research questions relevant to the contrasts you set up in part (i).
- (g). Below provided are some R outputs. Use them to verify the calculations you did in part (ii). Clearly state which output and which interval you used to verify your calculations.

```
k1<-c(0.5,0.5,-0.5,-0.5)
k2<-c(1,-1,0,0)
k3<-c(0,0,1,-1)

pupmEm<-emmeans(pumpFit,"design",data=wpump) #Marginal means
lincombs<-contrast(pupmEm,list(k1,k2,k3))
```

```
#Output A
confint(lincombs,adjust="none")
```

```
## contrast estimate SE df lower.CL upper.CL
## c(0.5, 0.5, -0.5, -0.5) 2719 107 44 2504 2934
## c(1, -1, 0, 0) 6038 151 44 5734 6343
## c(0, 0, 1, -1) -2389 151 44 -2694 -2085
##
## Confidence level used: 0.95
```

```
#Output B
confint(lincombs,adjust="bonferroni")
```

```
## contrast estimate SE df lower.CL upper.CL
## c(0.5, 0.5, -0.5, -0.5) 2719 107 44 2454 2985
## c(1, -1, 0, 0) 6038 151 44 5663 6414
## c(0, 0, 1, -1) -2389 151 44 -2765 -2014
##
## Confidence level used: 0.95
## Conf-level adjustment: bonferroni method for 3 estimates
```

```
#Output C
PostHocTest(pumpFit,method="bonferroni",conf.level=0.95)
```

```
##
## Posthoc multiple comparisons of means : Bonferroni
## 95% family-wise confidence level
##
## $design
## diff lwr.ci upr.ci pval
## D2-D1 -6038.4921 -6455.403 -5621.5814 < 2e-16 ***
## D3-D1 -6933.0419 -7349.953 -6516.1312 < 2e-16 ***
## D4-D1 -4543.6244 -4960.535 -4126.7137 < 2e-16 ***
## D3-D2 -894.5498 -1311.460 -477.6391 2.6e-06 ***
## D4-D2 1494.8677 1077.957 1911.7784 5.4e-12 ***
## D4-D3 2389.4175 1972.507 2806.3282 < 2e-16 ***
##
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```